



Knitted Textile Canopies: A Rapid Generative Design Optimization Tool for Enhanced Shading Performance

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Abstract

This study examines a rapid generative tool for adaptable lightweight shell structures to enhance shading in outdoor spaces. Addressing the escalating impacts of climate change and the urban heat island effect, the research explores innovative approaches to optimize shading knitted textile canopies. The study focuses on hot arid regions due to the limited shading provisions used in urban spaces in these regions, investigating maximum shadow coverage, performance enhancement, and material efficiency. With cutting-edge generative design tools, the method evaluates the effectiveness of various generative design plugins, such as Octopus, Galapagos, and Biomorpher, in generating efficient shading for canopies. The findings reveal that Biomorpher outperformed other plugins, achieving the highest shadow coverage in the shortest time. The optimum design was modified on a real site at the British University in Egypt campus, with adjustments to the structure and materials. Future research should incorporate additional generative tools, particularly those assessing thermal performance and structure stability, such as Finite Element Analysis, particularly for hybrid tensile membrane structures, to further enhance urban shading solutions and their resilience.

Keywords Generative design · Urban mitigation · Optimization · Knitted Textiles · Membrane · Shading efficiency

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Introduction

Climate change has rapidly increased the urban heat island effect, especially in hot arid climates, which has become one of the major concerns in highlighting resilience, community sustainable urban planning and adaptation strategies, especially in developing countries (Dhar and Khirfan 2017; Shahfahad et al. 2022). Green infrastructure has been recognized as a nature-based mitigation solution that can reduce temperature, provide cooler spaces, and enhance thermal comfort (Zhou et al. 2024), based on the canopy cover, volume, and density (Zhou et al. 2024). The lack of vegetation and shaded areas in outdoor spaces makes them increasingly vulnerable to extreme heat exposure, which negatively affects the users' comfort (Guo et al. 2023; Ou and Lin, 2023). Lightweight structures like canopies and shells play a vital role in improving urban microclimate by reducing direct solar radiation and enhancing temperature while providing essential shading (Nicholson et al. 2024). For instance, in Cairo, summer temperatures reach up to 50 °C, while shaded areas typically experience around 10 degrees lower, offering more comfortable conditions. Tree canopies can reduce the temperature from the generated shadows by 3.5–7.7 °C (Guo et al. 2023). The materials of these structures are crucial, not only in affecting the temperature underneath due to heat absorption or reflection, but also for recyclability, minimize material usage, less transportation, and the ease of assembly. For instance, textiles have been introduced to architecture as lightweight materials that can generate wide-span canopies with contemporary techniques, including hybrid structures and knitting membranes, achieving complex geometries, seamless three-dimensional shapes in a fast and waste-free manner (Anishchenko, 2021). Their inherent flexibility, durability, and aesthetic appeal enable the creation of dynamic, functional, and visually striking coverings that provide shade, weather protection, and spatial definition without adding excessive weight to the overall structure, unlike traditional techniques that consume much time to reach similar structures.

As building technologies rapidly evolve, addressing the potential of urban canopy design configurations in solving microclimate-level challenges requires generative design tools for addressing adaptation challenges to facilitate airflow and optimizing sun access (Gaber et al. 2020). Numerous studies have explored generative design alternatives and tools to facilitate the design process (Caetano et al. 2020; Jiang et al. 2022; Srivastava and Kawakami, 2023; Ahlquist et al. 2013; Van Ameijde et al. 2022; Sayed, 2001; Zhou et al. 2018; Vierlinger, 2013). Rubio's (2016) emphasised biomimicry as a conceptual approach to maximize shading efficiency and achieve optimal geometry forms (Davis-Sikora et al. 2020; Schwinn et al. 2013; Sayed, 2001).

Elrefai and Nikolopoulou (2023) conducted a study evaluating the outdoor shading structures as decision-making tools for selecting shade types, application period, design, and overall efficiency (Elrefai and Nikolopoulou, 2023). By using Ladybug Tools to calculate the Universal Thermal Climate Index, they assessed thermal comfort in different climates, comparing Cairo's hot environment

with London's milder climate. The findings indicated that Cairo requires high shading for about 2 h daily over 171 and 250 days, whereas London only needs shading for similar comfort levels on 26 and 45 days, highlighting the significant influence of climate on shading demands (Elrefai and Nikolopoulou, 2023). Nicholson et al. (2024) presented a novel workflow that simulates the impact of lightweight shade materials on outdoor thermal comfort by considering both shortwave and longwave effects. Unlike the standard method in Ladybug Tool, this approach incorporates shade interactions in the exchange of longwave radiation. The findings highlight the significant impact of shade material, height, floor material, and time of day on cooling effects, offering real insights for shade design optimization to enhance outdoor thermal comfort (Nicholson et al. 2024).

The implications of using generative design for lightweight shading structures warrant further investigation. Therefore, this study focuses on addressing this gap by digitally exploring generative design tools for new shading iterations of existing buildings, not through typology optimization. The aim is to find the best generative design tool for creating outdoor shading canopies that maximize shadow coverage while minimizing built-up area and material usage. Various optimization tools are tested during the design process, followed by comparison and evaluation of these tools. Thus, generative design should be considered during the design process of shading structures, which would lead to a better understanding of shading and the urban context.

Background

Knitted Membrane Textile Structural Applications

Shading structures can be constructed from various traditional materials, including wooden frames, plastic, fabric, and steel sheets. However, materials like wood and steel often face limitations in providing long coverage spans due to their reliance on post-and-lintel construction techniques. In contrast, tensioned fabric materials offer versatile forms and wide coverage capabilities, achieving complex architectural structures with minimal material usage (Hörteborn et al. 2023). They are known for their lightweight materials, foldable, and easy to transport, making them a practical option for temporary or semi-permanent installations (Nicholson et al. 2024).

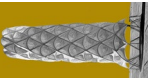
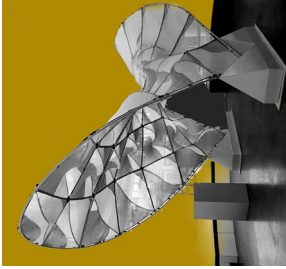
Generative design software can prioritize material efficiency while maximizing performance through shape optimization and material placement to achieve structural stability with minimal waste. Popescu et al. (2021) emphasized the advantages of lightweight fabric formwork systems as an alternative to conventional rigid formwork. These systems use flexible membranes to produce lightweight, waste-reducing formworks suitable for complex architectural components. Their portability and small size further enhance their practicality for construction sites. Kamal (2020) stated that membrane constructions, including both the supporting framework and tensioned fabric, can achieve spans ranging from 3 to 200 m (Kamal, 2020). The principles between generative design and fabric structure creation center on behaviour, adaptability,

efficiency, and innovative material use for novel solutions. For instance, parametric flexibility in the generative design enables the creation of adaptive complex forms that can respond to environmental factors like wind, light, and load distribution. Thus, traditional materials are being replaced with flexible and recyclable materials combined with hybrid structures during shell construction, enhancing sustainability and performance.

Table 1 shows an example of an innovative tensile membrane, (Popescu et al. 2021) referenced the NEST HiLo roof at ETH Zurich as a recent success in large-scale thin concrete shell construction. This project utilized a tensioned cable-net and woven fabric formwork to achieve a lightweight formwork, efficient structure, exemplifying the potential of fabric membranes in architectural innovation. Isoropia canopy for Denmark's 2018 Venice Biennale was constructed using a hybrid structure that merges both bent glass fiber reinforced polymer (GFRP) rods and knitted fabric (La Magna et al. 2018) that was CNC knitted across a cable-net fixed across the rods. Similarly, Hybrid Tower of 6.90 m high in Denmark was designed using generative design, combining bending active compression members (GFRP rods) with tensile members (knitted membrane) (Thomsen et al. 2015; Deleuran et al. 2015). Physical tests are conducted during textile rod assembly to assess behavior, revealing differences between simulated and actual structures caused by wind forces, which induce some bending at the sides. Knit Patterned Flow Pavilion used programmable CNC knitting machines to generate a knitted membrane, giving the three-dimensional volume to the structure (Tan, 2021; Tan et al. 2022). Using flexible rods inserted diagonally into the surface allowed the knitted panels to bend, forming the pavilion. Around 35 uniquely shaped, doubly curved panels are mounted on a bent aluminum boundary frame using 3D-printed joints and GFRP rods. The skin of the pavilion employs CNC-knit tailored membranes from glass fiber, enabling localized variations in elasticity within each textile (Anishchenko, 2021). Self-shaping textiles explored by Kycia through applying 3D printing techniques onto pretensioned fabric before release, enabling the fabric to morph into predetermined shapes autonomously and produce volumetric, dynamic, and adaptive surface structures that respond to internal or external stimuli (Kycia, 2022). This work aims to expand the possibilities of architectural textiles, offering new avenues for deployable, adaptive, and aesthetically dynamic surface structures.

Responding to environmental factors, generative design allows for the incorporation of environmental data, such as sun paths, wind flow, and thermal comfort, during the design process. Fabric structures can interact with environmental conditions to enhance their functionality and efficiency in urban contexts where they can serve as pavilions, canopies, or shelters, addressing issues like shading, heat mitigation, and public space enhancement. These structures can be tailored to fit specific site constraints and needs through algorithmic design processes. Most of these projects have shared generative design strategies and typology optimization to reach the optimum geometry, structures, and materials with less time while offering context-specific alternatives.

Table 1 Textile hybrid pavilion application

Pavillon	NEST HiLo roof ETH Zurich	Isoropia canopy CITA	Hybrid Tower CITA	Knit Patterned Flow Pavilion Dynamic Assemblies Lab
Illustration	Here supposed there was a pic 	Here supposed there was a pic ? 	Here supposed there was a pic ? 	Here supposed there was a pic ? 
Image credit	Michael Lyrenmann 2017	Anders Ingvarstsen 2018	Anders Ingvarstsen2015	LEE Tat Lin & TAN Ying Yi, 2020
Enhanced	By the authors			
Rods	Cable-net	(GFRP) rods	GFRP rods	3D-printed joints and GFRP rods
Surface	Woven fabric formwork	CNC knitted fabric across a cable-net	Knitted membranes	Knitted membrane

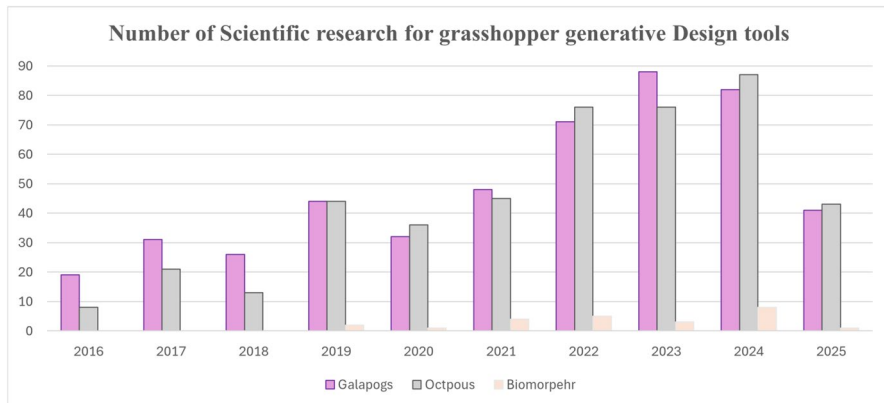


Fig. 1 Number of Scientific research for grasshopper generative design tools

Generative Design and Optimization Tool

Generative design is an algorithmic-driven design approach using a rule-based process involving a wide range of design options by iteratively generating and evaluating solutions based on predefined goals and constraints (Caetano et al. 2020). It often generates a new optimized form with less material usage and cost (Srivastava and Kawakami, 2023). In contrast, typology optimization is a mathematical technique that focuses on refining and optimizing existing designs based on specific parameters such as loads and density to improve the geometry's performance, efficiency, or suitability. While generative design emphasizes creative exploration and innovation through computational processes, typology optimization concentrates on enhancing and fine-tuning conventional or predefined design frameworks. Generative design has many digital design tools that include multi-objective optimization, which usually refers to solving problems with more than one competing objective, as defined by Liu et al. (2024). This method is widely applied in architectural design, where trade-offs between objectives often arise. Common optimization techniques include particle swarm optimization, simulated annealing, genetic algorithms, and surrogate model optimization. Early studies on single-objective focused on residential and office buildings, while multi-objective optimizations have been widely applied to educational buildings (Liu et al. 2024).

Grasshopper, a plugin for Rhino, offers tools such as Galapagos, Octopus, Wallace, Silvereye, Opossum, and Goat, which architects frequently use in design processes. Magdy et al. (2024) highlighted that hybrid simulations using Rhino–Grasshopper provide an effective way to maximize the potential of these plugins. Figure 1 shows the three highest tools published in the last ten years, which the research will focus on. To address the research gap in the application of Grasshopper tools, three plugins were selected for this study, while the other single-objective tools were excluded due to their limited suitability for this research's objectives. The selected plugins are.

- Galapagos: A single-objective optimization tool that serves as the reference point for comparison.
- Octopus: A multi-objective tool chosen for its ability to handle complex scenarios.
- Biomorpher: A less commonly used multi-objective tool, that provides a contrasting perspective.

Although Galapagos is a single objective tool, yet it has been mentioned in the published paper around 482 times which will be used as a reference point to the performance of other tools. Galapagos is a native Grasshopper component designed for single-objective evolutionary optimization, using either a genetic algorithm or simulated annealing (Vierlinger, 2013). Guo et al. (2022) highlighted its straightforward process, quick calculations, and simple convergence, making it an efficient tool for optimization. (Magdy et al. 2024) mentioned its versatility, citing its application in real-world projects such as the *Underwood Pavilion* (Łatka and Świeciak, 2021). Octopus, an external Grasshopper component, is tailored for multi-variable optimization. It employs an evolutionary algorithm inspired by natural selection, allowing designers to set user-defined criteria and design areas for geometry changes (Vierlinger, 2013). Magdy et al. (2024) described Octopus as a tool that integrates evolutionary algorithms with Pareto frontier solution sets to optimize multiple objectives. However, Chaichi and Garmaroodi (2024) noted its limited customization options for hyperparameters and algorithms. Biomorpher is an interactive design evolution tool within Grasshopper, as described by Vierlinger (2013). Bhavsar et al. (2024) utilized Biomorpher to optimize residential urban shapes at the neighborhood level, demonstrating its multi-objective optimization capabilities. A design for a fabric shading structure canopy will be evaluated based on these three plugins trying to find the most suitable one that can achieve the sharing principles between generative design and fabric structures listed up.

State of the Art

Based on the existing literature, several research gaps have been identified in the application of generative design for lightweight static shading structures. The implications of using generative design tools specifically for lightweight outdoor shading structure canopies have not been sufficiently explored to achieve sustainability and material efficiency in a short time. Additionally, there is a need for studies that specifically focus on optimizing multiple factors (e.g., maximizing shadow coverage and minimizing material usage) for shading structures using generative design approaches to mitigate the different changes that occur in the built environment on the urban scale. While various generative design tools are available, there is a lack of comprehensive studies comparing their effectiveness in optimizing shading structures for parameters such as shadow coverage, material efficiency, and structural performance.

The current scope of the research focuses on creating an efficient single shading element for covering large areas compared to multiple columns, tents, or metal

sheets used for shading areas. However, the research presents a road map for an effective shading structure that can be generated based on context requirements that change from one site to another through training the tool to set a goal based on input parameters. The paper assesses the tools' practical applicability, effectiveness and comparative performance in optimizing complex freeform canopies for efficient shading design, which are challenging to parameterize and optimize using traditional methods. This approach provides insights into the tools' potential limitations in real-world architectural applications.

The research question explores an experimental comparison that contributes to understanding the capabilities of three generative design tools in supporting innovative architectural design processes. Understanding the strengths and limitations of each tool can guide designers to select appropriate software for specific applications. Therefore, the study seeks to address these gaps by identifying and testing the best generative design tool that can generate computational outdoor shading canopies forms leading to more optimal, adaptable, and efficient shadows with the optimal utilization of materials.

Method

After exploring the different computational design types and highlighting generative design tools, the research aims to focus on exploring generative design tools to design outdoor shading structures with maximum shadow and minimum built-up area. To achieve this objective, the research method is divided into three main phases Fig. 2: (1) Basic Modeling (2) Case study factors and goals (3) Comparative analysis.

In this study, the comparison of tools such as Galapagos, Octopus and Biomorpher was based on their specific functionalities and ease of integration within our workflow, aiming to evaluate practical applicability rather than solely algorithmic performance. A basic shading element in the shape of a rectangle will be formed and deconstructed to its main structure of points and edges, giving values for a simple design (Phase 1), then by using the ladybug plugin, the cast shadow from the form will be studied along with the number and sum of area of anchor points of the structure (Phase 2), after that each generative design tool were given the needed data to run giving their results (Phase 3) which were analyzed and compared in results section.

Phase (1): Basic Modelling

This phase includes the modeling of the form created on Rhinoceros software to be modified by the generative tools. The geometry consists of a basic rectangle with a dimension of 10 * 5 m with a value height between 2.5 and 4.2 m. This area aimed to cover the students' gathering area at our campus where Fig. 4 shows the area's deconstruction into UV points which were later controlled by their X, Y, and Z values. Although the Corner points will have changes in their

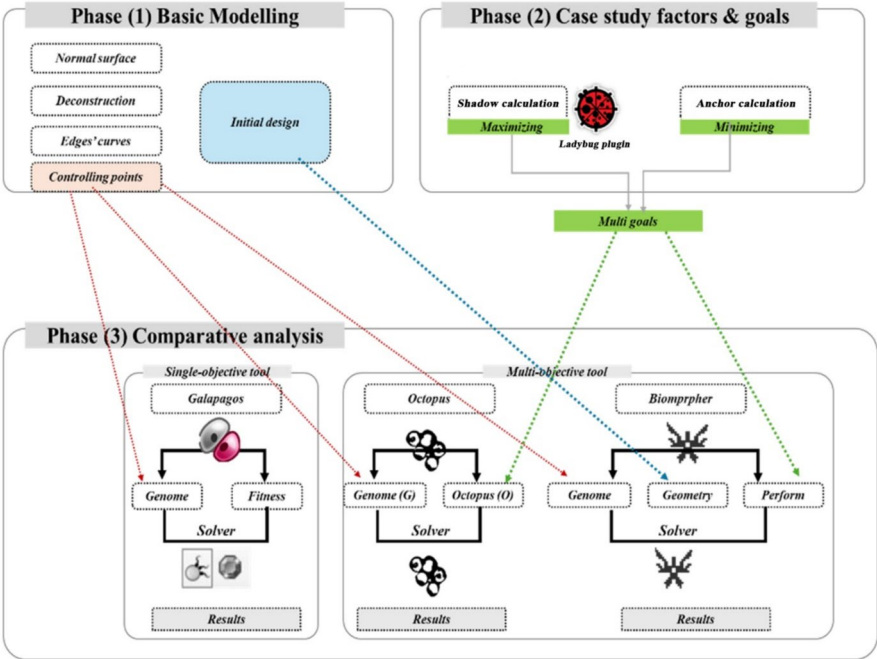


Fig. 2 Research workflow

Z values (from 2.5 to 4.2 m), the body points will have change in their X,Y,Z Values which leads to 33 sliders was patched to the main geometry to control the values of these points to reform the final shape. The final shape proceeded from the patch which was evaluated based on the percentage of shadows cast based on its area (Fig. 3).

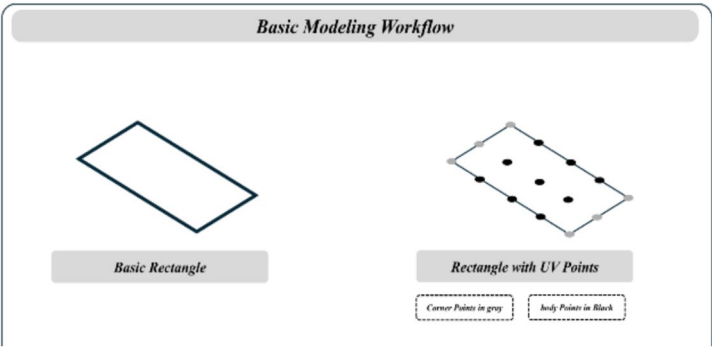


Fig. 3 Basic rectangle for generative design. Image: Rhino Viewport

Phase (2): Case Study Factors and Goals

Shadow Calculation Using the Ladybug Plugin

To determine the factors influencing the cast shadow, the Ladybug plugin within Grasshopper, running on Rhinoceros, was employed. This plugin utilizes 'epw' data, which is based on the selected location in Cairo, Egypt, to obtain thermal and sunlight information. Given that the sun's angles vary throughout the day and across different months, the analysis was set on August 21st, the warmest day of the year. To capture daily variations, three specific times were chosen for analysis: 9:00 a.m., 12:00 p.m., and 2:00 p.m. Using the data provided by the Ladybug plugin, the three outputs represented the outlines of the cast shadows at these times. Ultimately, the study's first goal was to calculate the ratio between the area of the cast shadow at each time and the built-up area, providing an initial measure of shadow coverage efficiency.

Anchor Calculation

By projecting studying surface points into the ground level (where $z=0$) and choosing the closest points to the center point of the surface, the locations of the anchors where the shed area will be fixed were set. This method determines the anchor location based on the closest set of points that had the minimum distance between the points and the center point. When connected to these points, these anchors will form a shape that acts as its vertices, and the area of this shape will later be used to calculate the minimum built-up area required for the canopy.

Phase (3): Comparative Analysis

In this phase, three generative design tools were tested to determine the suitable plugins for generative design by assessing their efficiency and capability in reaching the set goal of generating an outdoor shading structure for both Single and Multi-objectives optimization tools. By comparing both technical and practical assets of the plugins such as the amount of generated form in set time, CPU usage, the relevance of the generated form to the goal, and more. Generative design tools can be classified into 2 main categories based on the objectives they can deal with and solve. The first category is the single objective tools such as Galapagos and Ameba. While the second category is the multi-objective tools such as Octopus, Biomorpher, and Wallace. The main difference in the workflow is that the single objective tool can only deal with one single goal, which in our case is the generation of a shading surface that gives the maximum shadow at 2:00 p.m. In contrast, multi-objectives tools can deal with multiple goals in several time slots at once during the process of generating design as follows:

Goal (1): Generate a shading surface that gives maximum shadow at 9:00 a.m.

Goal (2): Generate a shading surface that gives maximum shadow at 12:00 p.m.

Goal (3): Generate a shading surface that gives maximum shadow at 2:00 p.m.

Goal (4): Generate an anchor system that has a minimum area.

Three different times were selected that represent the sun heat that affects daily life; 9:00 a.m., 12:00 p.m., and 2:00 p.m. The shadow percentage was calculated by dividing the area of the shadow over the area of the land. Therefore, the shadow percentage in some cases was above 100% due to the nature of saddle tensile structures. This study focused on three plugins for the experimentation, which are Galapagos as a single objective tool, Octopus, which is a widely used multi-objectives tool, and Biomorpher which is a new tool with a user-friendly interface that is considered an advanced Galapagos tool that uses multiple goals.

Application using the Galapagos Plugin

Galapagos is a plugin inside Grasshopper; it contains two inputs: Genome and Fitness. Genome is responsible for the sliders' number, which makes the Slover, while Fitness is responsible for the factor that wants to be maximized or minimized. By connecting all number sliders in the UV Curve (X, Y, Z) coordinates, as changes in their values would change the curves construct the Surface. After that, a single goal was chosen due to the nature of a single objective in the Galapagos, which was the shadow percentage at 2:00 pm.

Application Using the Octopus Plugin

The Octopus tool has four inputs; Phenoty, Phenotyp, Genome (G), and Octopus (O).

Both Phenoty and Phenotyp would not be used in this process, while Genome—like Galapagos will have all the number sliders of X, Y, and Z values. Unfortunately, Octopus only minimizes its goals, and the input only takes (number) as shown in Fig. 4. The area of shadows in the three different slots had a negative value and

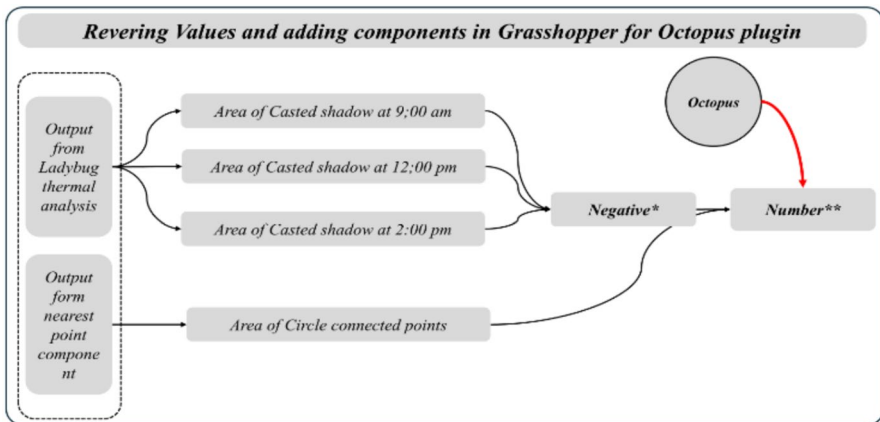


Fig. 4 Reversing values and numbering them to match Octopus Goal

number, but the radius output would only have the number as it was the target to be minimized.

Application using the Biomorpher Plugin

Biomorpher has 3 inputs: ‘Genome,’ ‘Geometry,’ and ‘Perform,’ unlike Galapagos, Biomorpher is a multi-objective tool that can achieve more than one goal in the same simulation step as Octopus. By connecting the three axes sliders (x, y, and z) with Values and Degrees of Curves to the Genome,’ initial design to Geometry input, and the area from shadow at the three times and radius into Perform, then, Biomorpher starts generating different designs to achieve the four goals mentioned earlier.

Results and Discussion

This section shows the comparative analysis results of each plugin from the number of meshes, area cast by shadow, time, efficiency, accuracy, and computing process to achieve the main research goal as shown in Fig. 5.

Results from Galapagos

The results from Galapagos showed around 38 genomes, the annealing solver as shown in Fig. 6 was used to give faster results to meet the single objective which was selected at 2:00 pm to calculate the cast of the shadow percentage.

A block of 15 min was set for the study to test each plugin. In the case of Galapagos, 142 meshes recorded their values. The data was exported to Microsoft

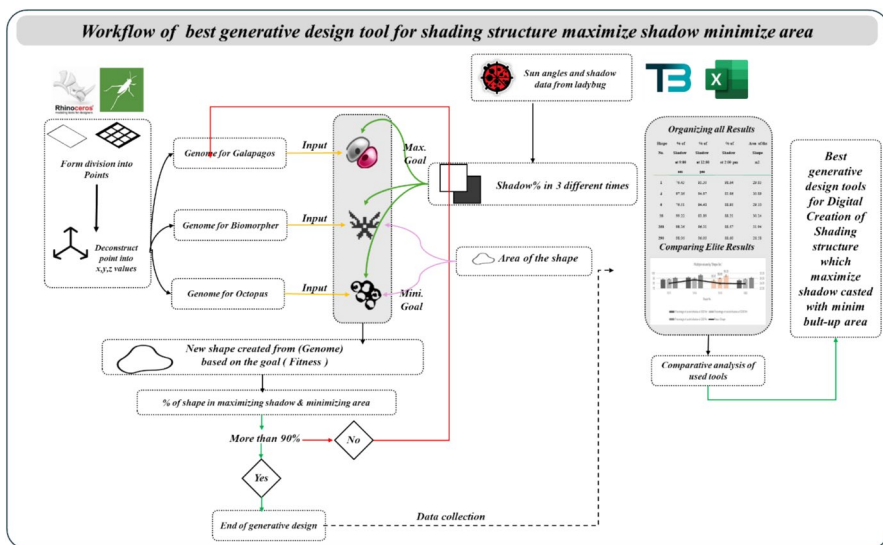


Fig. 5 Workflow for the generative design process in grasshopper tools

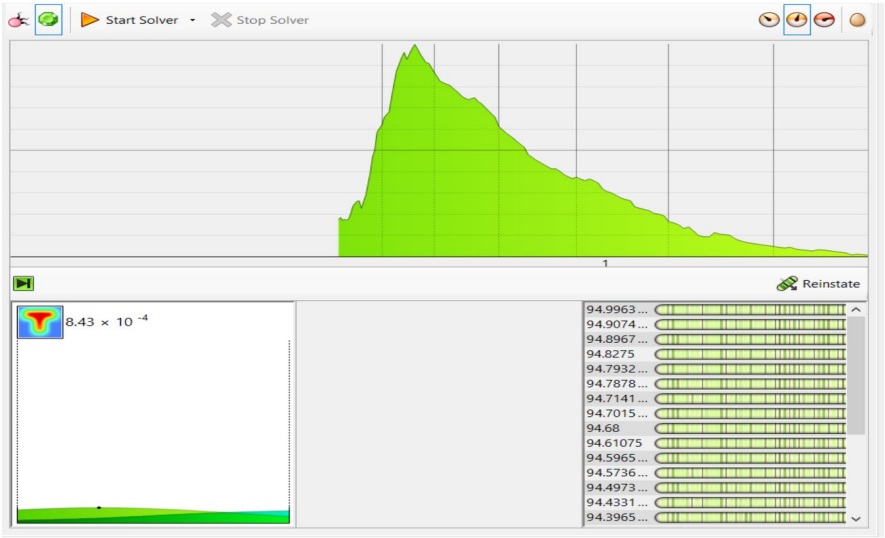


Fig. 6 The annealing analysis in Galapagos

Excel and the top 10% results shown in light green were highlighted to find the best solutions. It is observed that the area of the shadow at 2 pm is usually larger than the land area due to the sun’s angle, which increases the length of the shadow. The optimum 14 shapes were selected to calculate the area of each Fig. 7. The shading elements are intended to represent roof or canopy surfaces as overhead canopies which can be placed between two buildings or supported by columns. In addition,

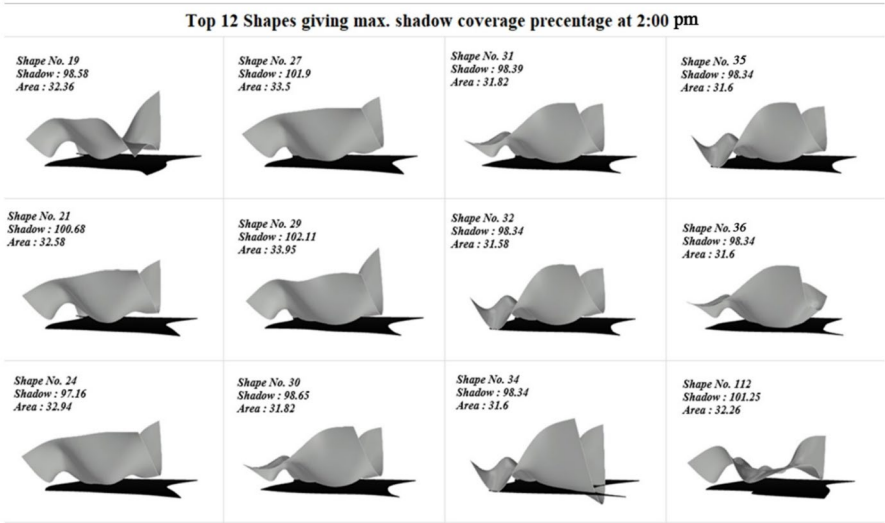


Fig. 7 Top 12 shapes giving maximum shadow at 2:00 pm by Galapagos

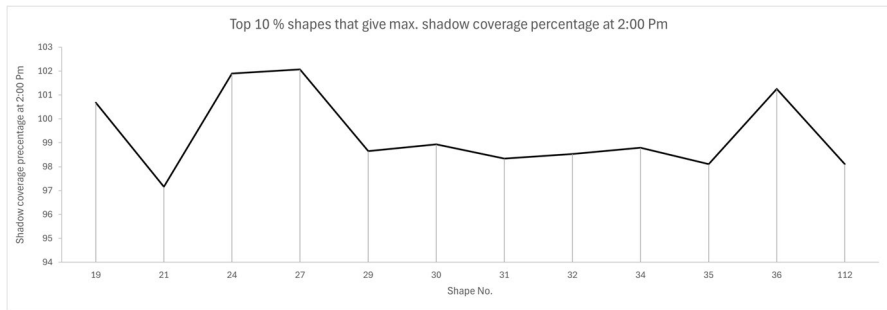


Fig. 8 Linear study of shadow coverage percentage and shapes by Galapagos

Table 2 Highlighting area values of each shape to determine the second goal

Shape No	Shadow coverage percentage at 2:00 p.m.	Area of Shapes (m ²)
19	98.58	32.36
21	100.68	32.58
24	97.16	32.94
27	101.9	33.5
29	102.11	33.95
30	98.65	31.82
31	98.93	31.82
32	98.34	31.58
34	98.53	31.6
35	98.79	31.6
36	98.11	31.27
112	101.25	32.26

“Anchor calculation” mentioned the anchor calculation based on the distance between the corners and the center point of the canopy, which can be fixed directly to the floor. The calculation was manually done in Grasshopper and exported to Excel as presented in a Linear Curve in Fig. 8 which visually represents the wide range of values that Galapagos can achieve in a single run. For instance, shape number 27 has the highest shadow coverage percentage, while its area value isn’t the smallest. Shape 36 has the smallest area but doesn’t achieve the maximum shadow coverage percentage.

By comparing all the data in (Table 2) and Fig. 9, it was observed that the maximum the shadows are, the larger the area is. Therefore, the optimal solution for shape no. 32 was chosen as the best shading structure from the Galapagos, highlighted in orange.

The working sequence in Galapagos shown in Fig. 10 involves a structured four-step process for optimizing parametric designs using Grasshopper. The workflow commences with Step 1, wherein the genome, representing adjustable parameters

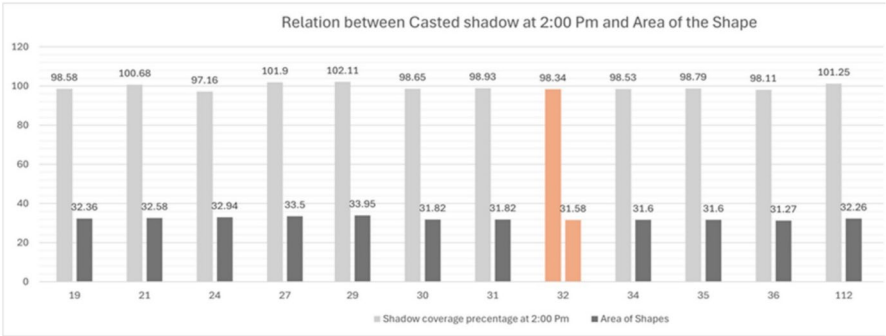


Fig. 9 Elite determination of the best solution for all geometry

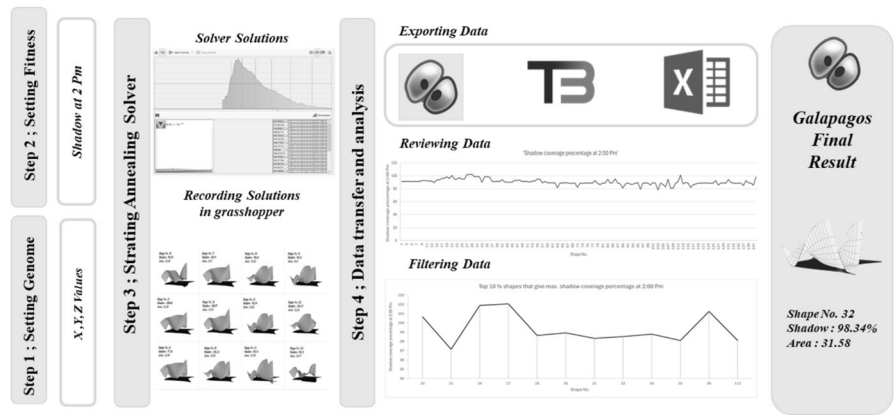


Fig. 10 Workflow of Galapagos

such as XYZ values, is established to define the input variables for optimization. Step 2 formulates the fitness function, typically based on predetermined performance criteria, such as minimizing or maximizing shadow coverage at specific temporal points (e.g., 2 pm). In Step 3, the optimization solver (e.g., Simulated Annealing or Evolutionary Solver) is initiated, enabling Galapagos to generate multiple solutions iteratively while recording and visualizing these in Grasshopper. This step yields graphical representations of potential outcomes and their performance. Subsequently, Step 4 focuses on data analysis and filtering, wherein results are exported (e.g., to Excel or other platforms) for comprehensive review. Visualization techniques, such as line graphs and bar charts, facilitate the identification of the most efficient design solutions. The process culminates in the identification of the optimal result, as exemplified in the given instance, where Shape No. 32 achieved a shadow coverage of 98.34% with an area of 31.58. This systematic approach ensures robust data-driven design decisions.

Results from Biomorpher

Although Biomorpher is considered a multi-objective Galapagos, it requires inserting the input data of geometry. Accordingly, after connecting all 38 genomes, the four goals, and the geometry, the solver started with its presets to generate the first generation (generation 0) shown in Fig. 11 in a single minute giving 12 possible solutions.

By changing the values of the design located on the left corner by setting shadows at 9 a.m., 12 p.m., and 2 p.m. to maximum and area to minimum as shown in Fig. 9, then evolving for five generations, Biomorpher provides 313 solutions in just 5 min, responding to the four goals. One of the key advantages of using Biomorpher is that it provides a history to track the evaluation of the solution thinking process, as it shows samples of each generation solution with their values and generation. Moreover, it provides a visual representation of the data and the relation between each generation as shown in Fig. 12, where each point represents a generated mesh (solution) and the highlighted dots are the ones chosen by the designer to be evolved, connected by lines to see the relations between the generations of the multiple designs. It visualizes the performance history of the generated canopy, specifically tracking the best values achieved in each generation. It depicts data points across the three selected times with the total area for each. Connecting the highest performance values across generations forms a series of lines that illustrate the trend of optimal performance over time. It shows that the performance metrics fluctuate across generations, where the area at 12:00 p.m. demonstrates peaks in certain generations, indicating areas of improved performance. The dotted scatter points represent individual data points within each cluster, providing a detailed overview of the distribution, while the connecting lines highlight the progression of the best-performing solutions in the evolutionary process. Moreover, the graph in Fig. 11 provides a representation of the relation between any 2 goals, creating

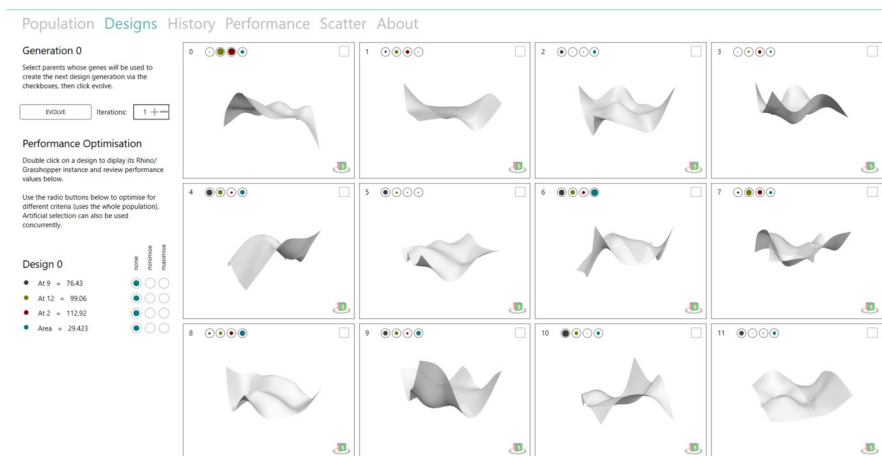


Fig. 11 Biomorpher generation 0 Designs tap

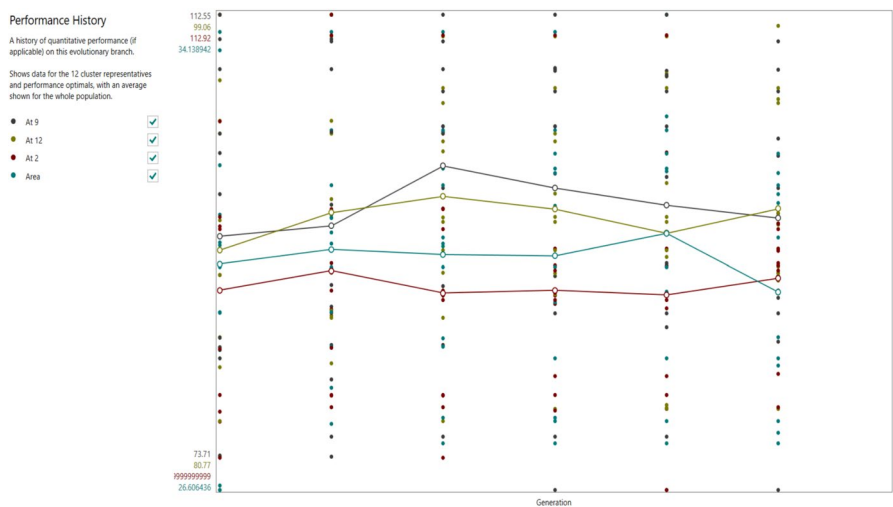


Fig. 12 Biomorpher linear study of generation performance by connecting the best value in each generation

a relationship between the cast shadow at 12:00 p.m. and the canopy area which each dot represents mesh in the selected generation. However, Biomorpher did not provide any written data that could help in the selection process of the optimal solution; that’s why Microsoft Excel was needed.

Figure 13 visualizes the relationship between two performance goals at 12:00 p.m. represented in the (x-axis) and the area represented in the (y-axis)—for a biomorpher-generated canopy. Each point represents an individual solution from different generations, with the black dots indicating higher-performing solutions and the lighter dots representing less optimal ones. The plot shows a wide distribution of solutions across the goal spectrum, highlighting trade-offs and correlations between

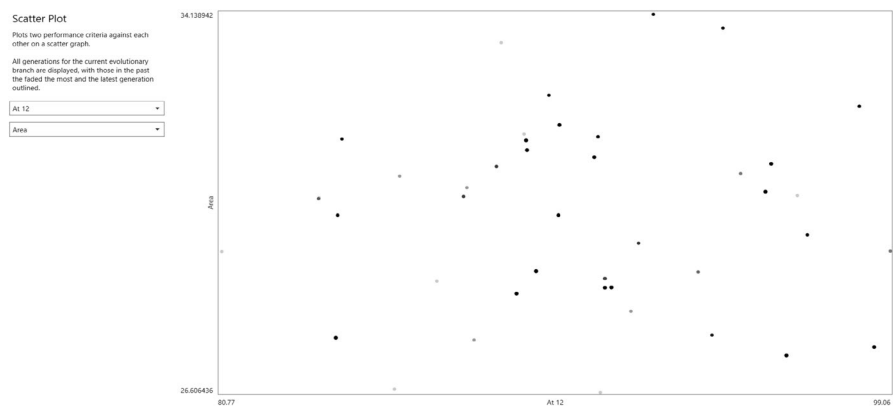


Fig. 13 Scatter Plot to show the relation between two goals in Biomorpher at 12:00 p.m.

the goals. The top right area contains solutions with high values in both goals, while the bottom left indicates lower-performance combinations. The chart effectively demonstrates the diversity of generated designs and how different solutions balance the two criteria, providing insight into the evolutionary process of optimizing the canopy design.

Instead of reviewing each goal independently, the top 10% values of each goal were selected and then combined to get the highest effective 30 shapes that at least have a top 10% value in a single goal (Table 3). The optimized geometry is no. 258 where it shows the less area and the maximum shadows. The visual comparison in

Table 3 Top 10% of shapes generated by Biomorpher in Excel analysis

Shape No	% of Shadow at 9:00 am	% of Shadow at 12:00 pm	% of Shadow at 2:00 pm	Area of the Shape m ²
1	76.43	95.50	98.64	29.85
4	97.86	94.97	83.96	30.89
6	76.51	94.48	88.95	29.10
20	97.86	94.97	83.96	35.57
21	86.72	85.49	78.41	26.61
27	83.44	88.57	93.58	29.33
28	99.22	83.89	88.21	30.24
34	76.51	94.48	88.95	26.84
60	94.43	92.45	96.89	28.88
65	90.46	94.48	88.95	26.61
75	85.54	88.84	92.37	33.86
81	98.01	89.99	79.86	30.14
84	85.46	88.57	93.58	27.99
93	97.58	85.64	90.90	28.23
97	86.99	89.95	91.41	33.86
99	98.36	94.28	94.78	28.81
136	86.99	89.95	91.41	28.88
140	90.39	89.10	90.67	27.76
166	73.29	89.95	91.41	27.05
208	99.54	87.39	79.82	29.77
218	74.43	94.48	88.95	30.51
219	98.36	84.21	91.02	29.58
220	78.08	95.64	94.27	27.76
234	78.16	96.09	88.60	33.39
239	85.84	91.07	88.12	27.35
243	99.25	91.26	81.54	32.30
258	98.36	96.31	88.57	31.94
290	98.96	96.09	88.60	28.58
301	98.36	95.80	94.73	27.05
306	96.29	83.89	91.11	28.23

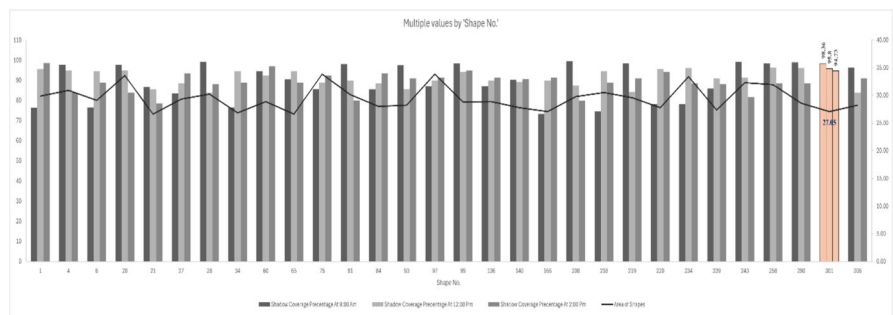


Fig. 14 Study of shadows and tradeline of the area of the Elite shapes by Biomorpher for optimum solution (orange)



Fig. 15 Biomorpher workflow

Fig. 14 shows that shape number 301 was common in all goals, which means that its values are the best in each goal, but it was the single shape to meet all four goals with high values.

The Biomorpher workflow in Grasshopper shown in Fig. 15 is a straightforward yet powerful process for exploring and optimizing design solutions. It starts with defining the input parameters, or genome, such as XYZ values, which serve as the foundation for the design. Next, the geometry is set up, along with the fitness criteria including factors like shadow coverage at different times of the day (9 a.m., 12 p.m., and 2 p.m.) with the total area calculation. After that, the Biomorpher solver is launched to generate different variations of design options. These options are displayed visually, making it easy to compare their performance. The data is exported to Excel for further analysis. The final step involves reviewing and analyzing the results in detail using graphs and charts to identify trends and the best-performing designs. Shape No. 301 emerged as the best solution, with shadow percentage coverage of 98.36%, 95.80%, and 94.73% at 9 a.m., 12 p.m., and 2 p.m.

respectively along with a total area of 27.05m^2 . This structured process makes it easy to explore design possibilities to make the best decisions.

Results from Octopus

Using all Octopus solvers for 15 min, more than 1070 results were created and viewed in different ways. As Octopus had both Biomorpher Visual representations of values in its multi-axes view and 3DWorld in Fig. 16a each cube represents a solution and it's located by the values of each goal, (b) the Multi-Axis Diagram where each line represents a solution, and it intersects with the Axis in its values. As mentioned in Sect. 3.3.2 Octopus always minimizes the goals and a negative number was assigned to the values in the Grasshopper code, therefore all the results are negative.

After exporting data to Microsoft Excel, and following the same analysis steps in Biomorpher, unfortunately, no shape was detected to have values more than 10%. Therefore, the range was expanded to cover an extra area of up to 20%, however, no shape was found to achieve the goal. Therefore, a criterion was set where any shape must cover an area of 20% or more and two maximum shadows at any two times represent the optimal outcome (Table 4). Following this criterion, only four shapes met these goals, with shape 1318 being the best possible scenario represented in Fig. 17.

The Octopus workflow in Grasshopper shown in Fig. 18 offers a clear and practical approach to multi-objective optimization. The process starts with defining the input parameters, or genome, such as XYZ values, which shape the design. Afterward, Octopus is set up by establishing goals, for example, maximizing shadow coverage at 9 a.m., 12 p.m., and 2 p.m., as well as minimizing the area of the shape. Once the objectives are in place, the Octopus solver is run, generating

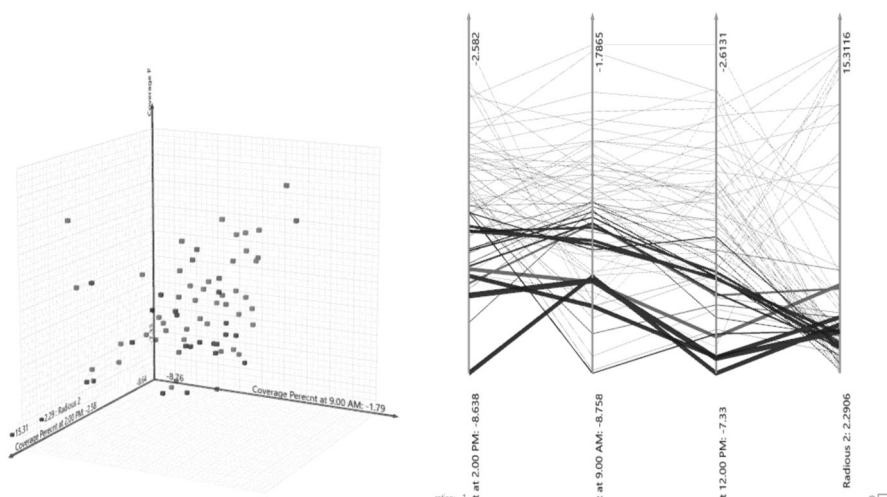


Fig. 16 a 3D World view in Octopus, b Multi-axes view in Octopus

Table 4 Sample of Top20% Shapes generated by Octopus in Excel analysis

Shape No	Shadow percentage at 9:00 am	Shadow percentage at 12:00 pm	Shadow percentage at 2:00 pm	Area of the shape m2
1313	90.81	85.51	85.66	24.25
1314	81.43	85.88	83.25	24.53
1315	85.34	85.48	88.52	24.04
1316	85.28	84.60	84.73	24.03
1317	83.24	84.59	84.72	26.39
1318	85.81	89.88	95.29	23.98
1319	85.77	85.48	85.65	23.99
1323	85.15	85.48	85.65	24.05
1324	84.05	84.66	84.92	25.30
1327	85.24	86.10	89.73	24.40
1328	85.65	85.22	85.32	25.19
1329	82.26	88.16	90.60	26.04
1330	85.14	86.68	90.54	24.04
1331	80.94	84.66	84.93	24.58

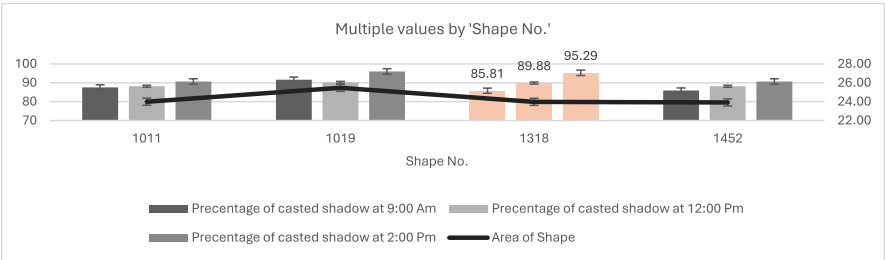


Fig. 17 Study of shadows and tradeline of Area of the Elite shapes by Octopus to determine the best solution (highlighted in orange) (1318)

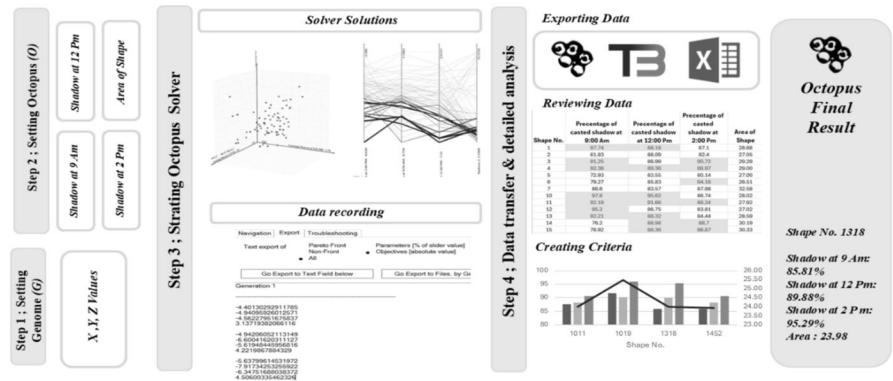


Fig. 18 Octopus workflow

and visualizing a range of design solutions in a multi-dimensional space. This visualization helps explore the trade-offs between different objectives. After the solver has completed, the data is recorded and exported to tools like Excel for deeper analysis. Charts and graphs are then used to review and compare the performance of the designs, making it easier to identify the most effective option. In this example, shape No. 1318 stood out as the best solution, achieving shadow coverage of 85.81% at 9 a.m., 89.88% at 12 p.m., and 95.29% at 2 p.m., with an area of 23.98. This process ensures a thorough exploration of possibilities and helps make well-informed design decisions.

Comparative Analysis of the Three Plugins

Table 5 shows the comparative analysis among the three plugins. Although Galapagos is a single objective tool, it did achieve all the required goals, unlike Octopus, which—unfortunately, despite being the most suitable tool in generating designs and the best in visualizing results in both texts and graphs, is the only one that could not achieve the three goals. On the other hand, Biomorpher plugin did well in generating a good number of alternatives in the shortest time at 1 min per generation—meaning 5 min in all—achieving all goals for both maximizing and minimizing factors. However, Biomorpher needs to provide the form as an input, making this less creative in predicting the generation technique. Based on this, it was observed that Biomorpher was the most efficient plugin in the case of the generation of lightweight structures for outdoor shading and generally better than Octopus on both input data systems and comparing results.

Constructive Approach of the Modified Canopy

Nature inherently optimizes structures by finding the most efficient paths, which often involve curvatures rather than straight lines. These forms can inspire canopy design structures that follow organic lines, reducing material use and enhancing stability, instead of forcing straight, rigid lines. For instance, a minimal surface with a curved frame, similar to a soap film stretched over a wire frame, has a minimal area due to surface tension. Inspired by this, self-shaping textiles follow a process where the main force acting on the structure is tension, compression, or bending. The proposed canopy construction employs a minimal surface principle inspired by the asymptotic Gridshell approach, geodesic curves (Gonzalez-Quintial and Martin-Pastor, 2021), similar to the Isoropia canopy (La Magna et al. 2018), combining both GFRP rods with knitted fabric for a lightweight yet stiff structure.

The canopy's innovative structure relies on a hybrid combination of materials, structures, and optimized geometry to achieve both lightness and strength Fig. 19. Based on the previous results, the optimized geometry was modified for a constructive and feasible structure that can be assembled as a self-supporting structure (El-Mahdy et al. 2025). Three anchors were fine-tuned and adjusted from the bottom to ensure they are positioned at the base for securing fixation to the ground.

Table 5 Comparative analysis of the outdoor shading between the 3 plugins' capabilities

	Galapagos	Biomorpher	Octopus
No. of goals	Single	Multi	Multi
Type of goals input	Integer output	Integer output	Number container only
Goals optimization	Maximizing or minimizing	Maximizing or minimizing	Minimizing only
Time	15 min	5 min for 5 generations	15 min
No. of Results*	142	313	1073
Recording of results	None	History of the Elite results in generations	History of all results
Data analysis	None	Scatter between 2 goals	None
Visual representation	None	Elite results	All results in Rhino Viewport
Achieving all goals	Yes	Yes	No
Shape properties	Shadow % at 9am	98.36	85.81
	Shadow % at 12pm	95.80	89.88
	Shadow at 2pm	94.73	95.29
	Shape Area	27.05	23.98

(*) based on running on 12 GB RAM with Intel Core i5 8th generation HP Laptop

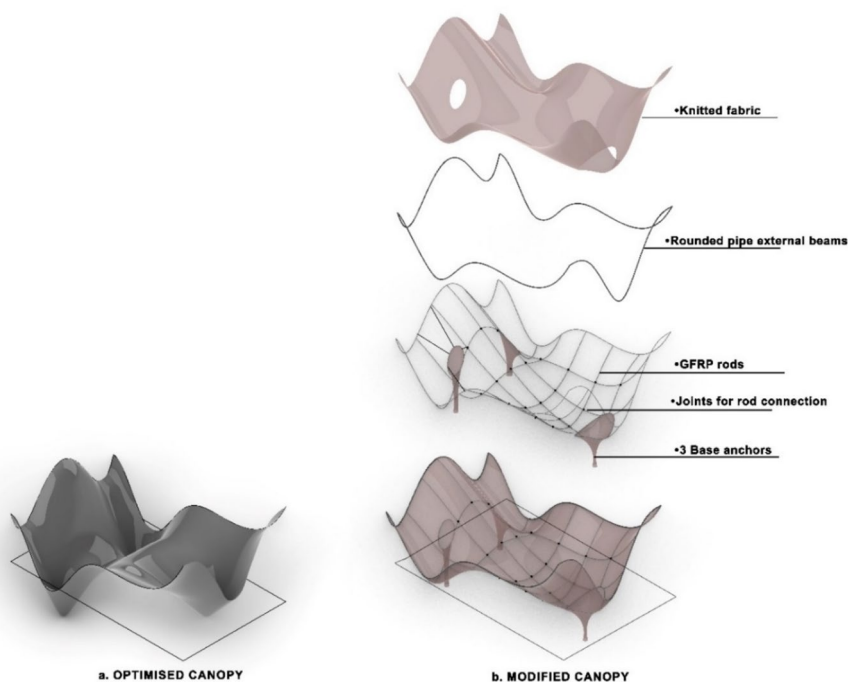


Fig. 19 The optimised and modified canopy with the construction approach installation

The structure includes bent GFRP rods distributed across the curvature of the shell, while external piped beams were proposed. The rods used as an alternative to traditional steel reinforcement, and valued for their high tensile strength, corrosion resistance, and lightweight, serve as tension elements. They are strategically positioned across the shell's curvature, supported by metal stoppers and anchored securely with adjustable fixtures. Inside, a cable net was installed on a waffle pattern running in both directions, providing additional support.

The skin of the materials' canopy is suggested to use CNC-knit fabric from PTFE-coated membranes, which is knitted to the structure via a cable net system that acts as a secondary reinforcement. While the knitted fabric and cable net distribute forces, forming a continuous, supportive membrane, creating a balanced system that efficiently withstands external forces like wind with a transparent and aesthetically appealing appearance, allowing minimal surface while ensuring structural integrity. This combination provides a feasible and visually striking approach that leverages the strengths of minimal surface geometry and suitable materials. The structure efficiently manages compression and tension forces through its interdependent components—GFRP rods handle tension, while the fabric and cable net provide compression and additional support.

Figure 20 shows the modified canopy installation in its context, where it is located between the library and building G. The canopy shows the different shadows in August, at three times; 9:00 a.m., 12:00 p.m., and 15:00 p.m.

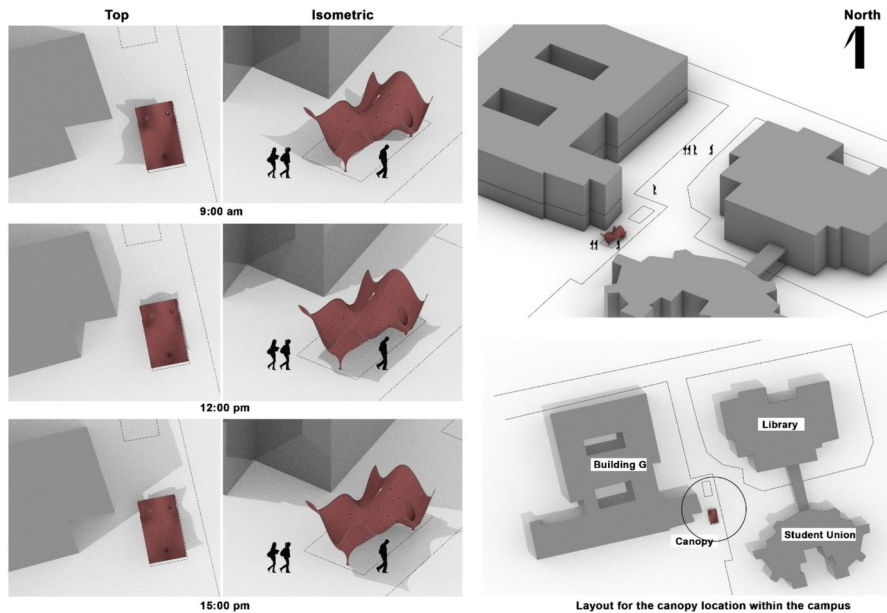


Fig. 20 Simulation for the shadow effect resulted from the canopy at three different times

Discussion

Some insights gained from the structure's performance could be integrated into urban planning strategies, contribute to microclimate modeling, and support sustainable design practices within cityscapes.

- The structural behavior of the shading element under various environmental conditions can influence microclimate factors such as wind flow, heat retention, and shading at the urban scale. For example, the findings of the optimized canopies could enhance the placement and design based on the building orientation to optimize natural ventilation or reduce urban heat island effects.
- The resulting simulation can give solution strategies on an urban planning scale by proposing design and structural modifications through adjusting the shade orientations to the building and optimizing the shading heights and spans which can increase shaded areas, improve thermal comfort, and enhance microclimate resilience. This would assist the urban planners in incorporating these insights into zoning codes, development guidelines, or master plans to create more climate-resilient cityscapes, ultimately fostering sustainable and comfortable urban environments.
- Different canopy typologies, such as single-layer, double-layer, or perforated designs, can significantly influence thermal comfort by affecting insulation, airflow, and shading. For example, a double-layer canopy with an air gap can improve insulation, reducing heat gain, while a perforated canopy can promote natural ventilation and cooling, similar to natural tree canopies. Incorporating

thermal comfort analysis in future research would provide a more holistic assessment of these designs' environmental performance and occupant comfort.

- Some potential pathways for future research that bridge the gap between structural analysis and urban environmental considerations can be suggested to integrate the simulation results with computational fluid dynamics models or urban climate simulations, to demonstrate the impact of the design on the wind direction based on the location and the exact environmental conditions. This would thereby strengthen the overall research contribution to architecture and urban microclimate studies.

Limitations and Further Studies

The selected tools succeeded in calculating the percentage of shadows at different times given the area of each, while the assessment of the structure calculations and load analysis in both the form of the fabric and the structure system were not covered in this research. Some structural plugins are left for further research to test the structure stability assessment, such as Kangaroo2, Ameba, Rhino Vault, and Wallace in addition to the Kiwi3d (Isogeometric Analysis) and SOFiSTiK (Finite Element Analysis) plugins. These plugins require high computing power during the analysis, which requires specific hardware PCs. Adding load analysis for the top 10–20% outcomes, whether inside the Rhino grasshopper environment or by using other CAD software this approach will close a gap and expand the knowledge base. While the current canopy shapes resemble free-form shell textiles, they may not fully capture the structural realities of tensioned fabric canopies, which rely on inner support elements like lattice shells, metal stoppers, cable-net, or tensioning systems to maintain their shape and maximize shade. Thus, incorporating form-finding tools such as Kangaroo could produce more physically accurate iterations and structurally feasible forms that resemble real-world fabric tensile structures. It would be efficient to build large scale models of vaulted textile membrane canopy for more bending active tensile membrane tests. This would be implemented on hybrid structures during the fabrication and assembly referenced to (La Magna et al. 2018). An advanced hybrid structure is also suggested for testing self-shaping textiles through 3D printing on pretension fabric before release, as already done by Kycia (2022).

The research presents a roadmap and workflow for any possible urban space to build an effective shading structure that can be used for many aspects based on context requirements. It concerns how to train the tool to set a goal based on parameters that change from one site to another. Shadow studies significantly impact thermal comfort, and this research aims to introduce new generative design methods, addressing a notable gap in shading structure research. By focusing on shadows initially, the study maintains simplicity, but future work could extend simulations to other aspects such as vegetation, fans, or water features using similar generative approaches to achieve sustainability and efficiency. For now, the emphasis remains on applying generative design to optimize shadow performance. The thermal comfort performance was not covered in this research; however, it could vary significantly across different canopy typologies.

Conclusion

Despite the ongoing evolution of design methods across various applications to address global challenges like climate change and pandemics, shading structure canopies continue to be constructed using traditional materials and techniques. Thus, by using different generative design tools in Grasshopper, this study showed a difference between single and multi-objective tools, how each one operates, and how to train them to set a goal based on parameters that change from site to another. Single objective tools like Galapagos provide easy usage and accurate results in a short time that can be studied and analyzed easily. Although Octopus provides a more generated design than other multi-objective tools like Biomorpher, it could not achieve the study goal in the set time. Therefore, this research proved that multi-objective tools like Biomorpher are suitable generative design tool to create a shading structure giving maximum shadow with the minimum built-up area due to its multi-objective nature, generation developing method in the design and accurate results that achieve all objectives assigned. Thus, it demonstrated a performance that was 200% better than Galapagos and 20% more efficient than Octopus. Additionally, it is time-efficient as it can compare over 300 designs in 5 min to generate a shading structure that could cast a shadow that covers around 90% of its assigned land at different times along the day with a small area and unique, double-curve saddle structure. This design ensures the canopy is lightweight, easy to assemble, and capable of withstanding external forces—an elegant blend of material science and geometric precision that exemplifies modern architectural innovation. Inspired by nature's minimal surfaces, self-shaping textiles open several innovative doors for shape generation depending on the main force acting on the structure, such as tension, compression, or bending. Future research into other generative design tools should focus on other applications related to people's comfort and design efficiency. Furthermore, while this paper tested only three generative design tools in Grasshopper, observational studies are required to gain more insight into different tools and applications, for instance, whether school windows are placed in a location that gives good amounts of lighting with less heat absorption of the sun using Wallace or Fusion 360, or are hospitals accommodation rooms are placed in the nearest location for the medical staff to have a quick reach for the patient.

Author contributions Mahmoud Hassan: writing – original draft, documentation, methodology, software, visualization, data collection, data analysis, results, discussion, resources. Esraa Torky: review & editing, editing and proofreading. Deena El-Mahdy: Conceptualization, supervision, writing – original draft, writing – review & editing, resources.

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Data availability Data will be made available on request.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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